

VISION2WEB

Full Stack Web Development Hackathon

Hackathon Problem Statements

Scope: Each challenge is intentionally designed to be achievable within 2 days by a student team. Participants are expected to build a working full-stack web application with a responsive frontend, backend APIs, database integration, validation, and a meaningful user workflow. Teams may use any suitable modern web stack.

General Expectations

- Responsive and user-friendly web interface.
- Frontend–backend integration through REST APIs or an equivalent approach.
- Persistent data storage using a relational or NoSQL database.
- Input validation and appropriate error handling.
- At least one role-based or workflow-based feature where relevant.
- A functional demo with sample data; advanced features are optional.
- Authentication, dashboards, search/filtering, analytics, or notifications should be implemented only where specified.

Theme 1: Smart Campus & Student Life

1. Campus Event Pulse

Build a web platform where students can discover campus events, view event details, register, and receive a registration status. Organizers should be able to create, edit, publish, and close events. Include category/date filters, limited-seat handling, a participant list, and a simple organizer dashboard showing registrations and available seats.

2. Lost & Found Hub

Develop a campus lost-and-found system where users can report lost or found items with images, location, date, category, and description. Implement search and filters, item status tracking, and a claim/request workflow. An administrator should be able to review reports and mark items as verified, claimed, or closed.

3. Skill Swap Network

Create a student skill-exchange platform where users can list skills they can teach and skills they want to learn. Users should be able to search profiles, send learning requests, accept or reject requests, and track active exchanges. Provide a dashboard showing offered skills, requested skills, and completed exchanges.

4. Campus Queue Manager

Build a digital queue management application for high-demand campus services such as certificates, help desks, or lab equipment booking. Users can join a service queue, receive a token number, view their approximate position, and cancel their request. Staff should be able to call the next token, pause a queue, and view basic queue statistics.

Core features: service queues, token generation, staff controls, live position, queue statistics.

5. Student Opportunity Tracker

Develop a portal that aggregates internships, workshops, hackathons, certifications, and competitions entered by an administrator. Students can browse opportunities, filter by type/deadline/skill, bookmark opportunities, and maintain an application status such as Interested, Applied, Shortlisted, or Completed. Include a personal dashboard with upcoming deadlines.

Theme 2: Community & Social Impact

1. Community Volunteer Match

Create a platform connecting volunteers with short-term community activities such as teaching, clean-up drives, donation collection, or awareness campaigns. Organizations can post opportunities with date, location, skills, and volunteer capacity. Users can apply, withdraw, and track participation. Include search/filtering and an organizer dashboard.

2. Neighborhood Issue Reporter

Build a web application for reporting local issues such as potholes, streetlight failures, waste accumulation, or water leakage. Users can submit a description, category, location, and photo. An authority/admin interface should assign status, priority, and remarks. Provide a dashboard showing issue counts by category and status.

Core features: issue submission, photo/location, admin status workflow, priority, dashboard.

3. Food Donation Coordinator

Develop a platform to coordinate surplus food donations from restaurants, events, hostels, or households with nearby NGOs/volunteers. Donors can post available food with quantity, pickup time, and location. Volunteers can claim pickups and update delivery status. Include expiry-aware filtering and a simple activity dashboard.

4. Emergency Resource Board

Build a community resource board for urgent non-medical support during local emergencies. Users can post requests for resources such as water, food packets, transport, or temporary shelter, while verified volunteers can respond to requests. Include request priority, status tracking, location-based filtering, and an admin verification workflow.

5. Accessibility-Friendly Local Services Directory

Create a directory of local services that records accessibility features such as wheelchair access, accessible restrooms, ramps, elevators, visual signage, and hearing assistance. Users can search and filter services based on required facilities and submit feedback. An admin should manage listings and moderate feedback.

Theme 3: Small Business & Productivity

1. Smart Inventory & Sales Dashboard

Build a lightweight inventory application for a small shop. Users can add products, update stock, record sales, and view low-stock items. The dashboard should display daily/weekly sales, top-selling products, and current stock levels. Include search, category filtering, and automatic stock reduction when a sale is recorded.

Core features: products, stock updates, sales entry, low-stock alerts, sales dashboard.

2. Service Booking & Slot Manager

Develop a booking system for a small service business such as a salon, repair center, tutor, or photography studio. Customers can view services and available slots, book appointments, and cancel them. Staff can define working hours, block unavailable slots, and view a daily appointment dashboard. Prevent double booking through backend validation.

Core features: services, availability, booking, cancellation, staff schedule, double-booking prevention.

3. Expense Split & Group Wallet

Create a group expense management application for trips, student teams, or shared activities. Users can create a group, record expenses, specify who paid and who benefited, and view simplified balances. Include settlement tracking, expense categories, and a dashboard summarizing total spending.

Core features: groups, expenses, participant split, balances, settlement tracking.

4. Invoice & Payment Tracker

Build a small-business invoice management system where users can create customers, generate invoices, add multiple line items, and update payment status. Include invoice search/filtering, overdue identification, and summary cards for paid, pending, and overdue amounts. A printable invoice view should be provided.

5. TaskFlow for Small Teams

Develop a collaborative task management application for small project teams. Users can create projects, assign tasks, set priorities and due dates, and move tasks through statuses such as To Do, In Progress, Review, and Done. Include comments, filtering, and a project dashboard with completion statistics.

Theme 4: Health, Wellness & Lifestyle

1. Personal Wellness Planner

Create a web application that allows users to set daily wellness goals such as water intake, exercise, sleep hours, or meditation. Users can record progress and view weekly trends. Include configurable goals, progress indicators, streak tracking, and a simple dashboard. The application should clearly state that it is for lifestyle tracking and not medical diagnosis.

2. Medicine Reminder & Family Schedule

Build a reminder application for managing medication schedules for family members. Users can create profiles, add medicines with dosage instructions and timing, and mark doses as taken or skipped. Provide a daily schedule and adherence summary. Include validation to avoid duplicate or conflicting entries.

3. Healthy Meal Planner

Develop a meal planning application where users can create a weekly meal plan from a predefined recipe database. Recipes should contain ingredients, preparation time, meal type, and basic nutritional information. Users can add meals to days, generate a consolidated shopping list, and filter recipes by preparation time or dietary preference.

4. Fitness Challenge Tracker

Create a platform for small groups to participate in fitness challenges such as walking, cycling, or exercise minutes. Users can join a challenge, enter daily progress, and view a leaderboard. Organizers can create challenges, define targets and dates, and monitor participation. Include validation for progress entries and a progress visualization.

Core features: challenge creation, participant progress, validation, leaderboard, progress chart.

5. Wellness Appointment Organizer

Build a scheduling system for non-emergency wellness services such as fitness consultations, yoga sessions, counseling appointments, or nutrition sessions. Users can browse available slots and book appointments, while service providers manage schedules and appointment status. Include reminders within the application, cancellation handling, and separate user/provider dashboards.

Theme 5: AI-Ready Web & Data-Driven Applications

1. Document Insight Dashboard

Build a web application where users can upload text-based documents and extract structured information such as title, keywords, word count, and user-defined tags. Store document metadata in a database and provide search/filtering. Optionally integrate an AI API to generate a short summary, but the core application must work without AI.

2. Smart Feedback Analyzer

Create a feedback management portal where users submit textual feedback and administrators view categorized results. Implement keyword-based or API-based sentiment/category analysis, store the analysis result, and provide charts for positive/neutral/negative feedback and common topics. Include feedback search and moderation.

Core features: feedback submission, classification/sentiment, moderation, analytics, search.

3. Resume Portfolio Builder

Develop a web application that allows users to enter profile, education, skills, projects, certifications, and experience details and automatically generates a responsive portfolio page. Provide multiple layout choices, profile preview, edit/update functionality, and a shareable public profile URL or identifier.

4. Visual Product Finder

Build a product catalog where users can search products using text filters such as category, price range, brand, and features. Add an optional image-upload feature that uses a simple AI/image API or predefined image tags to suggest visually related products. The application must provide a normal filter-based search as the reliable fallback.

5. Data Story Dashboard

Create an interactive dashboard from a provided dataset such as sales, transport, energy, attendance, or public-service data. Users should be able to upload or select a dataset, filter records, and view at least three meaningful visualizations. Include summary metrics, a searchable data table, and an export option for filtered results.

Core features: dataset handling, filters, KPIs, charts, searchable table, export.

Evaluation Areas

- Problem understanding and completeness of the solution
- Frontend usability and responsive design
- Backend/API design and database integration
- Functional correctness and validation
- Innovation and meaningful use of the chosen theme
- Code quality, security awareness, and error handling
- Demo quality and ability to explain technical decisions